

● HARD

● INFORMATION AND IDEAS

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03

10 questions · ~15 min

Q1

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The following text is adapted from María Cristina Mena's 1914 short story "The Vine-Leaf."

It is a saying in the capital of Mexico that Dr. Malsufrido carries more family secrets under his hat than any archbishop.

The doctor's hat is, appropriately enough, uncommonly capacious, rising very high, and sinking so low that it seems to be supported by his ears and eyebrows, and it has a furry look, as if it had been brushed the wrong way, which is perhaps what happens to it if it is ever brushed at all. When the doctor takes it off, the family secrets do not fly out like a flock of parrots, but remain nicely bottled up beneath a dome of old and highly polished ivory.

Based on the text, how do people in the capital of Mexico most likely regard Dr. Malsufrido?

- A. Many have come to tolerate him despite his disheveled appearance.
- B. Few feel concerned that he will divulge their confidences.
- C. Some dislike how freely he discusses his own family.
- D. Most would be unimpressed by him were it not for his professional expertise.

The ancient Greek concept of “mimesis,” a term used in the works of Plato, Aristotle, and other Greek philosophers in discussions of representational art—visual, performance, or literary art that aims to depict the real world—is a foundational concept of the Western philosophy of aesthetics. Mimesis is typically translated as “imitation” in modern editions of ancient Greek texts, but scholar Stephen Halliwell warns that this is overly reductive: “imitation” implies that art merely copies—and is thus by definition entirely derivative of—a reality that exists outside and prior to the work of art, and translating “mimesis” thusly obscures the multifaceted ways in which the ancient Greeks understood the relationship between art and reality.

Which statement, if true, would most directly support the claim by Halliwell presented in the text?

- A. One of the earliest appearances of mimesis’s root word, *mimos*, can be found in an ancient Greek tragedy in reference to dramatic impersonation, and the *mim*- root came to be generally associated with the musical and poetic arts by the fifth century BCE.
- B. Both Plato’s and Aristotle’s theorizations of mimesis examine the psychological effects that works of art induce in the viewer or listener.
- C. Although several of Plato’s earliest philosophical works discuss aesthetic ideas, the term “mimesis” doesn’t appear in Plato’s discussions of art until *Cratylus*, a relatively late work.
- D. Although Plato’s writings typically characterize representational art as an inferior reflection of the physical world, Aristotle suggests that mimesis can refer to art’s capacity to envision hypothetical conditions that could, but don’t yet, exist.

In a 2017 article, historian Jasmine Kilburn-Toppin explains that in early modern London, members of the city's guilds (trade and artisanal associations) were participants in a civic culture in which gift giving both signaled and conferred social status. Research on this phenomenon has tended to focus on philanthropic gifting by London's largest guilds; for her part, Kilburn-Toppin focuses on the gifting of handmade objects and fixtures (such as decorative paneling or plasterwork) within the craft guilds, which were "composed of highly discerning producers and consumers of material cultures." Given this characterization, it can reasonably be inferred that the gifting of such objects may have _____ blank

Which choice most logically completes the text?

- A. ensured that knowledge of the materials and techniques used by members of the craft guilds stayed within the craft guilds.
- B. emphasized ostentatious offerings in the case of the largest guilds and prosaic objects in the case of the craft guilds.
- C. functioned as a way for craft guild members to maintain and enhance their professional reputations among their peers.
- D. conferred greater social status when the recipient belonged to one of London's largest guilds than when the recipient belonged to a craft guild.

Paleontologist Lucas E. Fiorelli and colleagues have reported the discovery at a mine in Brazil of several egg clutches, partially preserved single eggs, and egg shells from the Late Cretaceous period. The researchers have concluded that the area was once a nesting and breeding site for titanosaurs, a group of sauropod dinosaurs. The finding is significant given the previous lack of known nesting sites in northern regions of South America, which led many paleontologists to assume that titanosaurs migrated south to lay eggs.

What does the text most strongly suggest about the site discovered by the researchers?

- A. It is the earliest known example of a titanosaur nesting and breeding site.
- B. It was very difficult to excavate given that it was discovered in a mine.
- C. It may have been occupied by other sauropods in addition to titanosaurs.
- D. It is farther north than any other nesting site discovered in South America.

Neural networks are computer models intended to reflect the organization of human brains and are often used in studies of brain function. According to an analysis of 11,000 such networks, Rylan Schaeffer and colleagues advise caution when drawing conclusions about brains from observations of neural networks. They found that when attempting to mimic grid cells (brain cells used in navigation), while 90% of the networks could accomplish navigation-related tasks, only about 10% of those exhibited any behaviors similar to those of grid cells. But even this approximation of grid-cell activity has less to do with similarity between the neural networks and biological brains than it does with the rules programmed into the networks.

Which finding, if true, would most directly support the claim in the underlined sentence?

- A. The rules that allow for networks to exhibit behaviors like those of grid cells have no equivalent in the function of biological brains.
- B. The networks that do not exhibit behaviors like those of grid cells were nonetheless programmed with rules that had proven useful in earlier neural-network studies.
- C. Neural networks can often accomplish tasks that biological brains do, but they are typically programmed with rules to model multiple types of brain cells simultaneously.
- D. Once a neural network is programmed, it is trained on certain tasks to see if it can independently arrive at processes that are similar to those performed by biological brains.

Conventional theories of rhetoric hold that presenting information as coming from credentialed experts increases that information's credibility. When communications researcher Sungkyoung Lee and her colleagues tested messages seeking volunteers for clinical trials, however, they found that participants in their study judged recruitment messages from former trial volunteers as significantly more credible than messages from doctors (i.e., credentialed experts). One reason for this may be that the doctors' status as credentialed experts wasn't ignored but rather was outweighed by participants' views of the experiential relevance of the two types of messengers; that is, participants may have reacted the way they did because _____ blank

Which choice most logically completes the text?

- A. messages from former trial volunteers depicted clinical trials as being more positive experiences than did messages from doctors.
- B. participants did not have enough experience to evaluate the credibility of the doctors' messages but did have enough experience to evaluate the credibility of former trial volunteers' messages.
- C. the fact that former trial volunteers went through the same experience that participants were contemplating while doctors did not was more important to participants than the doctors' status as credentialed experts was.
- D. participants regarded the experiences of both the doctors and former trial volunteers as relevant to the subject of clinical trials but were skeptical of the doctors' status as credentialed experts.

Philosophers note that many people have an intuitive sense that while we ought not to lie, there may be circumstances in which lying is permissible. If this intuition is correct and we lack an inviolable duty to speak truthfully, what grounds opposition to lying in the first place? Japa Pallikkathayil has advanced one answer by appealing to a duty to respect others' agential interests: the possession of false beliefs constrains agency, and thus we ought not to impede the formation of true beliefs unless doing so prevents a greater constraint on someone's agency or an otherwise impermissible end.

Which choice best states the main idea of the text?

- A. One potential means of justifying opposition to lying is Pallikkathayil's argument that we have an obligation to respect other people's agency that entails a commitment to truthfulness except in certain circumstances.
- B. Many people have an intuitive sense that lying is permissible in some circumstances but lack a principled way to identify those circumstances, and Pallikkathayil's argument may provide a means of resolving that problem.
- C. Pallikkathayil's argument suggests that if we have a duty to respect other people's agential interests and if possession of false beliefs constrains agency, then we have an inviolable duty to speak truthfully.
- D. Pallikkathayil's argument shows that if our intuition that circumstances may make lying permissible is correct, then it is unclear whether there are any grounds for an opposition to lying in the first place.

The Souls of Black Folk is a 1903 book by W.E.B. Du Bois. In the book, Du Bois suggests that upon hearing Black folk songs, he felt an intuitive and sometimes unexpected sense of cultural recognition: _____ blank

Which quotation from *The Souls of Black Folk* most effectively illustrates the claim?

- A. “[Black folk music] still remains as the singular spiritual heritage of the nation and the greatest gift of the Negro people.”
- B. “Ever since I was a child these songs have stirred me strangely. They came out of the South unknown to me, one by one, and yet at once I knew them as of me and of mine.”
- C. “Caricature has sought again to spoil the quaint beauty of the music, and has filled the air with many debased melodies which vulgar ears scarce know from the real. But the true Negro folk-song still lives in the hearts of those who have heard them truly sung and in the hearts of the Negro people.”
- D. “The songs are indeed the siftings of centuries; the music is far more ancient than the words, and in it we can trace here and there signs of development.”

Outi Tervo and team studied the effect of human-caused noise on narwhals (*Monodon monoceros*), arctic marine mammals that are sensitive to acoustic changes in their environment. Hypothesizing that elevated sound levels affect foraging among narwhals, Tervo’s team compared narwhal diving behaviors in natural sound conditions with those behaviors in two human-caused sound exposure conditions—ship sounds and ship sounds coupled with sonic pulses. Both exposure conditions resulted in significant decreases in the number and target depth of deep dives (associated with foraging) relative to natural conditions. However, differences between diving behaviors in the two exposure types were negligible, a finding that could be attributed to the fact that _____ blank

Which choice most logically completes the text?

- A. sonic pulses can be heard at significantly greater ocean depths than ship sounds can.
- B. ship sounds contribute so much to the overall sound level that the addition of sonic pulses has little effect on the narwhals’ auditory environment.
- C. narwhals forage at shallower depths in the presence of ship sounds alone than in the presence of ship sounds coupled with sonic pulses.
- D. the narwhals weren’t as sensitive to human-caused sounds as the researchers had predicted.

Optical tweezers are specialized scientific tools—particularly useful in biology and medicine—that use high-powered beams of light to trap and manipulate minuscule particles for study. Use of the tool has led to several scientific and medical breakthroughs over the last few decades, but the particles are often under prolonged exposure to the intense heat of the light beams. To overcome the risk of overheating, and thereby damage, researchers sometimes attach nano-sized glass beads to particles, allowing the light to focus on the beads instead of the particles.

Based on the text, what is one advantage of attaching glass beads to particles when using optical tweezers?

- A. It decreases the time it takes for the optical tweezers to locate and capture the particles.
- B. It facilitates the maneuvering of particles without directly heating the particles themselves.
- C. It allows researchers to use weaker light beams to manipulate particles.
- D. It adds a material to which particles can transfer any heat absorbed from the optical tweezers' light beam.